Experiment

1. Open the Binomial module in your web application.

opened binomial module

2. Enter n = 20 and p = 0.05.

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3. Calculate the mean, variance, and probability of exactly 2 defects.

mean = np = 20 x 0.05 = 1

variance = np(1-p) = 20 x 0.05 x 0.95 = 0.95

P(X = 2) = 0.1887

4. View the PMF graph. The horizontal axis represents the number of defects and the vertical axis represents the probability of each count.

highest bar at 1 defect (0.3774), then 0 defects (0.3585) and 2 defects (0.1887)

after 4 defects the bars are almost zero so graph is right skewed

5. Change the defect probability from 0.05 to 0.10. Observe how the distribution shifts toward larger defect counts.

with p = 0.10 mean = 2, variance = 1.80, P(X = 2) = 0.2852

peak moved from 1 to 2 defects so graph shifted right and got wider

6. Open the Poisson module and enter λ = 5 for 100 inspected products. Compare the Poisson probabilities with the Binomial results.

poisson λ = 5 : P(0) = 0.0067, P(2) = 0.0842, P(5) = 0.1755

binomial n = 100, p = 0.05 : P(0) = 0.0059, P(2) = 0.0812, P(5) = 0.1800

values are almost same so poisson can be used instead of binomial when n is big and p is small

7. Open the Geometric module and enter p = 0.05. Examine the probability of detecting the first defect on different inspection attempts.

1st attempt = 0.05, 2nd = 0.0475, 3rd = 0.0451, 5th = 0.0407, 10th = 0.0315, 20th = 0.0189

mean = 1/p = 20 inspections

probability keeps decreasing slowly with every attempt

8. Write down the numerical results, graph observations, and an inference about the quality-control process.

results: p = 0.05 gives mean 1, variance 0.95, P(X = 2) = 0.1887 and p = 0.10 gives mean 2, variance 1.80, P(X = 2) = 0.2852

graphs: binomial graph shifts right when p increases, poisson graph looks almost same as binomial, geometric graph keeps going down

inference: at 5% defect rate the process is under control (mostly 0 to 2 defects in 20), at 10% more defects come so the process needs checking, poisson works for big batches and on avg 20 inspections are needed to find the first defect